

Notes on lecture 18

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1 Oriented surfaces

Intuitively speaking, a surface having 2 distinct sides is called an **oriented surface**. That is, at each point on the surface, we should be able to draw 2 normal vectors, one inward normal and another outward normal.

Question: Why can't all surfaces be oriented surfaces?

Answer: All surfaces can't be oriented surfaces. An example of a non-oriented surface is a mobius strip. If we consider a rectangular strip with corners A, B, C, D (as shown the left hand side picture in Figure 1.0.1) and roll it to make a band (by joining the corner B with C , and A with D), then this surface will have 2 distinct sides. If an ant is at one point of that band like surface and the ant wants to reach the point that lies exactly at the backside of it, then the ant will have to cross the edge of the surface.

Instead of this, if we join the corner A with C and B with D , then we will

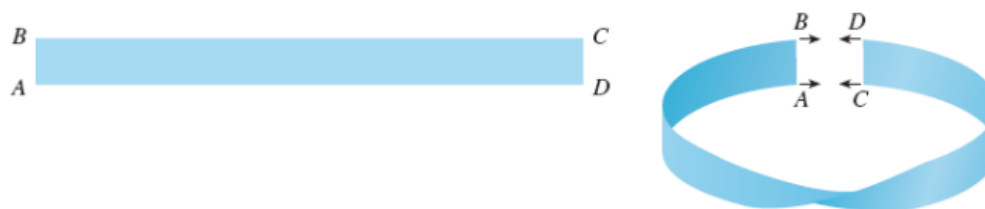


Figure 1.0.1: Construction of a mobius strip

get a mobius strip (as shown in the right hand side picture in Figure 1.0.1). Then an ant starting from one point of the surface can reach the backside point by traversing the surface without crossing any of its edges. Such a surface is NOT an oriented surface. This finishes the answer to our question.

From now on, we consider only oriented (two-sided) surfaces. We start with a surface S that has a tangent plane at every point (x, y, z) on it (except at boundary points). There are two unit normal vectors $\hat{n}_1$ and $\hat{n}_2 (= -\hat{n}_1)$ at (x, y, z) . Please see Figure 1.0.2 for this.

If it is possible to choose a unit normal vector $\hat{n}$ at every such point (x, y, z)

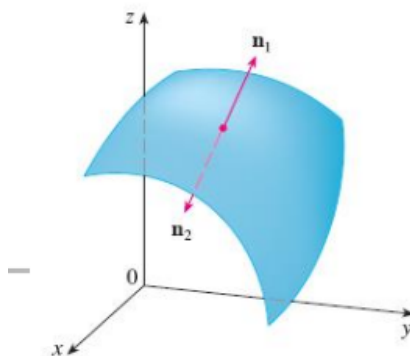


Figure 1.0.2: An oriented surface

so that $\hat{n}$ varies continuously over S , then S is called an **oriented surface** and the given choice of $\hat{n}$ provides S with an **orientation**. There are 2 possible orientations for any oriented surface. See Figure 1.0.3 for this.

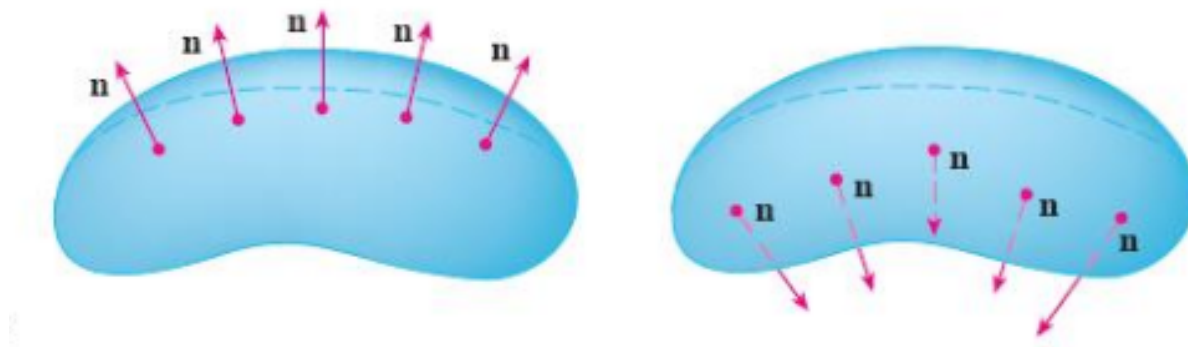


Figure 1.0.3: Two possible orientations for an oriented surface

Figure 1.0.4 shows a mobius strip, which is a non-oriented surface.

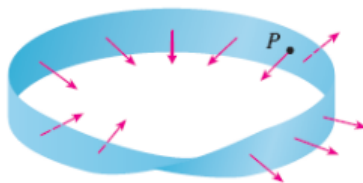


Figure 1.0.4: A mobius strip (non-oriented surface)

Any surface S with equation $z = g(x, y)$ can be regarded as a parametric surface with parametric equations

$$x = x, y = y, z = g(x, y).$$

So we have

$$\vec{r}_x = \hat{i} + g_x \hat{k} \text{ and } \vec{r}_y = \hat{j} + g_y \hat{k}.$$

Thus

$$\vec{r}_x \times \vec{r}_y = -g_x \hat{i} - g_y \hat{j} + \hat{k}. \quad (+ + +)$$

We use equation $(+ + +)$ to associate with the surface a natural orientation given by the unit normal vector

$$\hat{n} = \frac{-g_x \hat{i} - g_y \hat{j} + \hat{k}}{\sqrt{(g_x)^2 + (g_y)^2 + 1}}.$$

Since the $\hat{k}$ -component is positive, this gives the *upward* orientation of the surface.

If S is a smooth oriented surface given in parametric form by a vector function $\vec{r}(u, v)$, then it is automatically supplied with the orientation of the unit normal vector

$$\hat{n} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}$$

and the opposite orientation is given by $-\hat{n}$.

For a **closed surface**, that is, a surface that is the boundary of a solid region E , the convention is that the **positive orientation** is the one for which the normal vectors point *outward* from E , and inward-pointing normals give the negative orientation. Figure 1.0.5 shows positive orientation for a sphere. Similarly, Figure 1.0.6 shows negative orientation for a sphere.

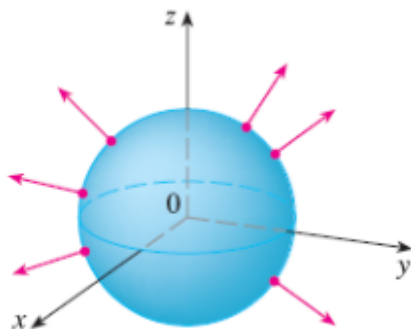


Figure 1.0.5: Positive orientation

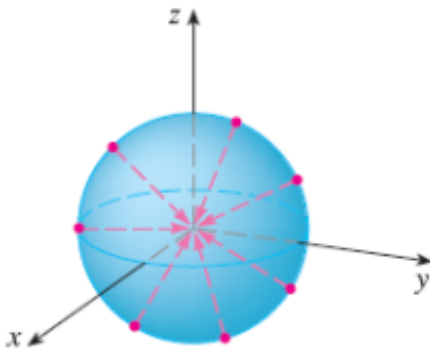


Figure 1.0.6: Negative orientation

All vectors on the surface of the sphere, which are going to point outwards of the sphere, are called outward normals. There is a similar notion for inward normals. In the lower hemisphere, the outward normals point downwards, and the inward normals point upwards.

1.1 An interesting example

Consider the sphere $x^2 + y^2 + z^2 = a^2$. Any point on the sphere can be expressed in parametric form as

$$\vec{r}(\phi, \theta) = a \sin \phi \cos \theta \hat{i} + a \sin \phi \sin \theta \hat{j} + a \cos \phi \hat{k}.$$

Take any point on the surface of the sphere and join it with the origin. Then we will get the radius vector $\vec{r}$. From the **exercise** below, it is easy to see that the radial vector is in the direction of the normal to the surface.

Exercise: Show that

$$\hat{n} = \frac{\vec{r}_\phi \times \vec{r}_\theta}{|\vec{r}_\phi \times \vec{r}_\theta|} = \frac{1}{a} \vec{r}(\phi, \theta).$$

The radial vector itself is normal to the surface of the sphere.

2 Surface integral of vector fields

Earlier, when we studied ordinary surface integrals, we computed $\int_S \int f(x, y, z) dS$ for some scalar function f over a surface S . Now, this scalar function $f(x, y, z)$ is going to be replaced by the dot product of some vector functions (which being a dot product, is again a scalar function).

We are going to consider a surface S and a point $P(x, y, z)$ on it. At the point P , we are going to consider an outward unit normal vector $\hat{n}$. Suppose we define a vector field $\vec{F}$ over the surface S (that is, at each point of S , we are able to define a vector-valued function $\vec{F}$), then the dot product $\vec{F} \cdot \hat{n}$ is going to be a scalar function and we are simply going to compute $\int_S \int \vec{F} \cdot \hat{n} dS$. That is, the integrand $f(x, y, z)$ in ordinary surface integrals is going to be replaced here by $\vec{F} \cdot \hat{n}$. Figure 2.0.1 illustrates this.

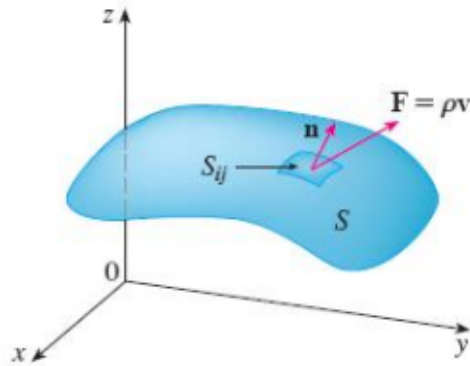


Figure 2.0.1: Surface integral of vector fields

For computing the surface integral of the vector field $\vec{F}$ over a surface S , we

are going to proceed as follows:

We divide the surface S into patches, and on each patch S_{ij} , we consider the product

$$(\vec{F} \cdot \hat{n})(x_{ij}^*, y_{ij}^*, z_{ij}^*)A(S_{ij}),$$

where $A(S_{ij})$ is the area of the patch S_{ij} and $(x_{ij}^*, y_{ij}^*, z_{ij}^*)$ is a point on S_{ij} . Then we take a double sum of the above expression over all i, j , and take limits as the number of patches tend to ∞ . That will give us the surface integral of the vector field $\vec{F}$ over S .

2.1 A surface immersed in a fluid

We are going to assume that the surface S is like an imaginary net which is immersed in some fluid and the fluid is flowing over that imaginary surface. Assume also that S is a closed surface. Consider a patch on the surface S . Then we will be able to draw an imaginary column inside the surface (as shown in Figure 2.1.1).

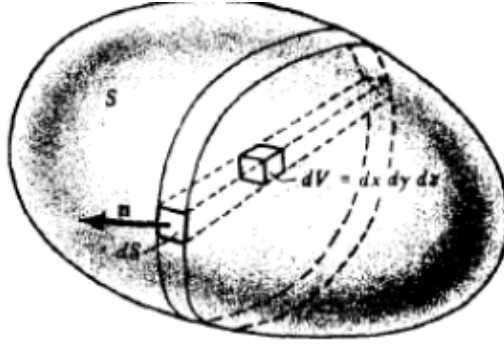


Figure 2.1.1: An imaginary column inside the surface S

Then the fluid is going to enter the surface through the patch (that we had considered), pass through the column, and it is going to come out of the column from the other side of the surface.

2.2 A digression

At this point, let us recall the concept of scalar projection. Suppose we want to consider the scalar projection of a vector $\vec{b}$ over another vector $\vec{a}$. Let θ be the angle between $\vec{a}$ and $\vec{b}$. Then if l denotes the scalar projection of $\vec{b}$ over $\vec{a}$, we have

$$\frac{l}{|\vec{b}|} = \cos \theta.$$

This implies that

$$l = |\vec{b}| \cos \theta = |\vec{b}| |\hat{a}| \cos \theta = \vec{b} \cdot \hat{a},$$

where $\hat{a}$ is the unit vector along $\vec{a}$.

Now suppose we are considering some vector field $\vec{F}$ and we want to compute the component of $\vec{F}$ along a particular direction $\vec{a}$, then that component is simply going to be given by $\vec{F} \cdot \hat{a}$, where $\hat{a}$ is the unit vector along $\vec{a}$.

2.3 Back to a surface immersed in a fluid

Let us once again consider the surface S immersed in a fluid. Suppose the fluid is moving with a velocity $\vec{v}$ and $\hat{n}$ is a unit normal vector at a point on the patch, then the component of $\vec{v}$ along the direction of $\hat{n}$ is given by $\vec{v} \cdot \hat{n}$. We know from physics that

$$distance = velocity \times time.$$

Now, if we assume that Δt is the time taken by the fluid to cross the column, then the length of the column will be given by $(\vec{v} \cdot \hat{n})\Delta t$. And hence the volume of the column is given by $\{(\vec{v} \cdot \hat{n})\Delta t\}dS$, where dS denotes the area of the patch.

We know from physics that

$$density = mass \text{ per unit volume}.$$

Then the mass of the fluid inside the column for time Δt is given by

$$\rho\{(\vec{v} \cdot \hat{n})\Delta t\}dS,$$

where ρ denotes the density of the fluid. That means mass of fluid contained in the column per unit time is going to be

$$\frac{\rho\{(\vec{v} \cdot \hat{n})\Delta t\}dS}{\Delta t} = \rho(\vec{v} \cdot \hat{n})dS.$$

If we perform the integration $\int_S \int \rho(\vec{v} \cdot \hat{n}) dS$, then that will give us the mass of the fluid contained inside the whole surface S per unit time.

That means, if we consider $\rho\vec{v}$ to be some vector valued function $\vec{F}$, then $\int_S \int \rho(\vec{v} \cdot \hat{n}) dS$ will represent something like $\int_S \int \vec{F} \cdot \hat{n} dS$.

Remark 2.3.1. The integral $\int_S \int \rho(\vec{v} \cdot \hat{n}) dS$ was nothing but the mass of the fluid going across the surface per unit time. Whenever we have some quantity moving across something per unit time, that quantity is called the **flux**.

Notation 2.3.2. $d\vec{S} = \hat{n} dS$. □

Definition 2.3.3. If $\vec{F}$ is a continuous vector field defined on an oriented surface S with unit normal vector $\hat{n}$, then the **surface integral of $\vec{F}$ over S** is

$$\int_S \int \vec{F} \cdot d\vec{S} = \int_S \int \vec{F} \cdot \hat{n} dS.$$

This integral is also called the **flux** of $\vec{F}$ across S . □

If

$$\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k}$$

is a parametrization of the surface S , then we know that

$$\hat{n} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}$$

and

$$dS = |\vec{r}_u \times \vec{r}_v| dA,$$

where dA is the area of the corresponding patch in the uv -plane [The patch in the uv -plane need not always be a rectangle].

Then we have

$$\begin{aligned} \int_S \int \vec{F} \cdot d\vec{S} &= \int_S \int \vec{F} \cdot \hat{n} dS \\ &= \int_D \int \vec{F} \cdot \frac{(\vec{r}_u \times \vec{r}_v)}{|\vec{r}_u \times \vec{r}_v|} |\vec{r}_u \times \vec{r}_v| dA = \int_D \int \vec{F} \cdot (\vec{r}_u \times \vec{r}_v) dA, \end{aligned}$$

where D is the parameter domain in the uv -plane for the surface S .

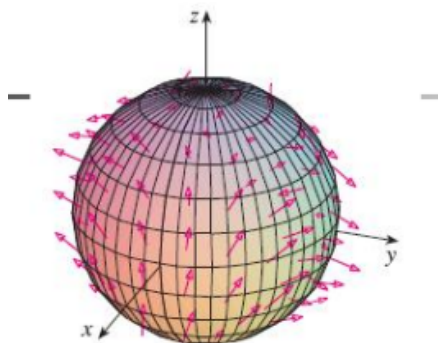


Figure 2.3.1: Figure for Example 2.3.4

Example 2.3.4. Find the flux of the vector field $\vec{F} = z\hat{i} + y\hat{j} + x\hat{k}$ across the unit sphere $x^2 + y^2 + z^2 = 1$. Figure 2.3.1 demonstrates the vector field $\vec{F}$ over the unit sphere.

The unit sphere can be represented parametrically as

$$\vec{r}(\phi, \theta) = x(\phi, \theta)\hat{i} + y(\phi, \theta)\hat{j} + z(\phi, \theta)\hat{k},$$

where

$$x(\phi, \theta) = \sin \phi \cos \theta, y(\phi, \theta) = \sin \phi \sin \theta, z(\phi, \theta) = \cos \phi$$

and

$$(\phi, \theta) \in D = [0, \pi] \times [0, 2\pi].$$

Then the required flux equals

$$\int_D \int \vec{F} \cdot (\vec{r}_\phi \times \vec{r}_\theta) dA,$$

where $D = [0, \pi] \times [0, 2\pi]$.

Recall from earlier lectures that

$$\vec{r}_\phi \times \vec{r}_\theta = \sin^2 \phi \cos \theta \hat{i} + \sin^2 \phi \sin \theta \hat{j} + \sin \phi \cos \phi \hat{k}.$$

Hence

$$\int_D \int \vec{F} \cdot (\vec{r}_\phi \times \vec{r}_\theta) dA = \int_{\phi=0}^{\pi} \int_{\theta=0}^{2\pi} [2 \sin^2 \phi \cos \phi \cos \theta + \sin^3 \phi \sin^2 \theta] d\phi d\theta.$$

The integral in the right hand side of the above equation can be broken up as the sum

$$\int_{\phi=0}^{\pi} \int_{\theta=0}^{2\pi} [2 \sin^2 \phi \cos \phi \cos \theta] d\phi d\theta + \int_{\phi=0}^{\pi} \int_{\theta=0}^{2\pi} [\sin^3 \phi \sin^2 \theta] d\phi d\theta$$

Observe that both the integrals in this sum are in variable separable form, so they can be easily calculated. In fact, the first integral will be equal to 0 because $\int_0^{2\pi} \cos \theta = 0$. The rest of the calculation is left to the reader as an exercise.

We can perform the same integration in some other way as well: Observe that $\hat{n}$ is the outward unit normal vector on the surface of the sphere and the sphere is given by $x^2 + y^2 + z^2 = 1$. We can consider the sphere as a level surface given by $\Phi(x, y, z) = 0$, where $\Phi(x, y, z) = x^2 + y^2 + z^2 - 1$. Then

$$\hat{n} = \frac{\nabla \Phi}{|\nabla \Phi|} = \frac{\langle 2x, 2y, 2z \rangle}{2\sqrt{x^2 + y^2 + z^2}} = \langle x, y, z \rangle,$$

where the last equality follows because $x^2 + y^2 + z^2 = 1$. Now

$$\int_S \int (\vec{F} \cdot \hat{n}) dS = \int_S \int (zx + y^2 + xz) dS = \int_S \int (2zx + y^2) dS.$$

You can easily perform the integration now, using the fact that

$$dS = |\vec{r}_\phi \times \vec{r}_\theta| dA.$$

□

Remark 2.3.5. If the surface S is the graph of a function of 2 variables (say, $z = g(x, y)$), and $\vec{F} = \langle P, Q, R \rangle$, then

$$\begin{aligned} \int_S \int (\vec{F} \cdot \hat{n}) dS &= \int_D \int \langle P, Q, R \rangle \cdot \frac{\langle -g_x, -g_y, 1 \rangle}{\sqrt{g_x^2 + g_y^2 + 1}} \sqrt{g_x^2 + g_y^2 + 1} dA \\ &= \int_D \int (-Pg_x - Qg_y + R) dA, \end{aligned}$$

where D is the shadow/projection of the surface S on the xy -plane.

Example 2.3.6. Find the flux of the vector field $\vec{F} = \langle y, x, z \rangle$ over the surface S where S is the boundary of the solid region E enclosed by the paraboloid $z = 1 - x^2 - y^2$ and the plane $z = 0$.

Here, we have to compute $\int_S \vec{F} \cdot d\vec{S}$, where S is as mentioned above. Observe that the surface S is the union of 2 surfaces S_1 and S_2 as shown in Figure 2.3.2.

Then

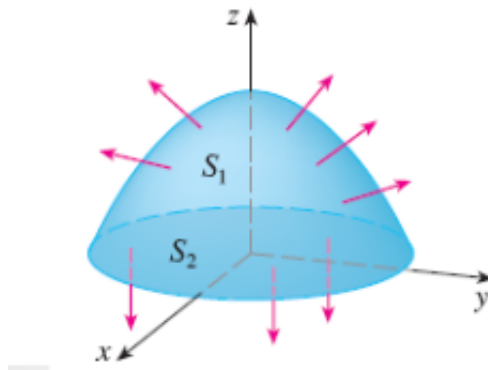


Figure 2.3.2: Figure for Example 2.3.6

$$\int_S \vec{F} \cdot d\vec{S} = \sum_{i=1}^2 \int_{S_i} \int \vec{F} \cdot d\vec{S} = \sum_{i=1}^2 \int_{S_i} \int (\vec{F} \cdot \hat{n}) dS.$$

Let us consider the surface S_2 first. The surface S_2 is the base disk, and therefore an outward unit normal to S_2 will point in the vertically downward direction. This direction is simply $-\hat{k}$. So on S_2 , we have:

$$\vec{F} \cdot \hat{n} = (y\hat{i} + x\hat{j} + z\hat{k}) \cdot (-\hat{k}) = -z.$$

But $z = 0$ on S_2 . Therefore we get that $\int_{S_2} \int (\vec{F} \cdot \hat{n}) dS = 0$.

Let us consider S_1 now. For S_1 , we have $z = 1 - x^2 - y^2$. So

$$\hat{n} = \frac{\langle -z_x, -z_y, 1 \rangle}{|\langle -z_x, -z_y, 1 \rangle|} = \frac{\langle -z_x, -z_y, 1 \rangle}{\sqrt{z_x^2 + z_y^2 + 1}}.$$

Here $z_x = -2x$ and $z_y = -2y$. It then follows from Remark 2.3.5 above that

$$\int_{S_1} \int (\vec{F} \cdot \hat{n}) dS = \int_D \int (2xy + 2xy + z) dA,$$

where D is the base disk S_2 . So

$$\int_{S_1} \int (\vec{F} \cdot \hat{n}) dS = \int_D \int (4xy + z) dA = \int_D \int (4xy + 1 - x^2 - y^2) dA,$$

where the last equality follows because $z = 1 - x^2 - y^2$. It is left to the reader as an exercise to complete the above integration using polar coordinates. $\square$

3 Stokes theorem

Recall the definition of an oriented surface first. An oriented surface has two distinct sides, and at any point on the surface, one can draw two normal vectors (one outwards and another inwards). Figure 3.0.1 shows an oriented surface S with unit normal vector $\hat{n}$.

The orientation of S induces the **positive orientation of the boundary**

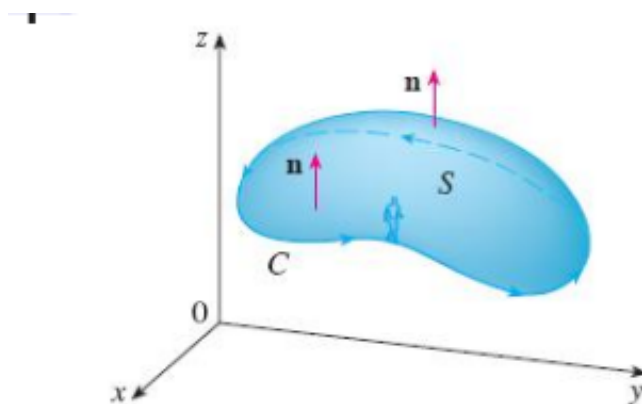


Figure 3.0.1: An oriented surface

curve C as shown in Figure 3.0.1. This means that if you walk in the positive direction around C with your head pointing in the direction of $\hat{n}$, then the surface will always be on your left.

Theorem 3.0.1. Stokes theorem: *Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let $\vec{F}$ be a vector field whose components have*

continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S . Then

$$\oint_C \vec{F} \cdot d\vec{r} = \int_S \int (\nabla \times \vec{F}) \cdot d\vec{S}.$$

Since

$$\oint_C \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot \vec{T} ds \text{ and } \int_S \int (\nabla \times \vec{F}) \cdot d\vec{S} = \int_S \int (\nabla \times \vec{F}) \cdot \hat{n} dS,$$

therefore Stokes theorem says that the line integral around the boundary curve of S of the tangential component of $\vec{F}$ is equal to the surface integral over S of the normal component of the curl of $\vec{F}$. In the equation above, s denotes the arc length and $\vec{T}$ is a unit tangent vector to the curve C .

Remark 3.0.2. In the special case, where the surface S is flat and lies in the xy -plane with upward orientation, the unit normal is $\hat{k}$. Then the surface integral becomes a double integral and Stokes theorem becomes

$$\oint_C \vec{F} \cdot d\vec{r} = \int_S \int (\nabla \times \vec{F}) \cdot d\vec{S} = \int_S \int (\nabla \times \vec{F}) \cdot \hat{k} dA.$$

This is precisely the vector form of Green's theorem. Thus we see that Green's theorem is really a special case of Stokes theorem.

Stokes theorem converts a line integral into a surface integral. Proving Stokes theorem for a general surface is extremely difficult. However, in this lecture, we will prove Stokes theorem for a special case, when the surface is the graph of a function of 2 variables.

PROOF: (of Stokes theorem in the special case): Suppose the surface S is given by $z = g(x, y)$. Let the projection of S over the xy -plane be the region D . Let S be bounded by the closed curve C . Let C_1 denote the corresponding curve in the xy -plane. Figure 3.0.2 provides an illustration of this.

If C_1 is parametrized by

$$\vec{r}'(t) = \langle x(t), y(t) \rangle, \quad a \leq t \leq b,$$

then C is parametrized by

$$\vec{R}(t) = \langle x(t), y(t), g(x(t), y(t)) \rangle, \quad a \leq t \leq b.$$

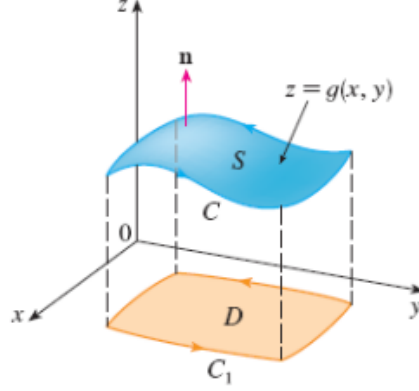


Figure 3.0.2: Proof of Stokes theorem in the special case

The right hand side integral (RHS) in the statement of Stokes theorem is given by:

$$RHS = \int_S \int (\nabla \times \vec{F}) \cdot d\vec{S} = \int_S \int \{(\nabla \times \vec{F}) \cdot \hat{n}\} dS.$$

If $\vec{F} = \langle P, Q, R \rangle$, then

$$\begin{aligned} RHS &= \int_S \int \{(\nabla \times \vec{F}) \cdot \hat{n}\} dS \\ &= \int_D \int \left\langle \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right\rangle \cdot \frac{\langle -g_x, -g_y, 1 \rangle}{\sqrt{g_x^2 + g_y^2 + 1}} \sqrt{g_x^2 + g_y^2 + 1} dA \\ &= \int_D \int \left\langle \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right\rangle \cdot \langle -g_x, -g_y, 1 \rangle dA \\ &= \int_D \int \left[-g_x \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) - g_y \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \right] dA. \end{aligned} \quad (*)$$

Let us now concentrate on the left hand side integral (LHS).

$$\begin{aligned} LHS &= \oint_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{R}(t)) \cdot \vec{R}'(t) dt \\ &= \int_a^b \left(P \frac{dx}{dt} + Q \frac{dy}{dt} + R \frac{dz}{dt} \right) dt. \end{aligned}$$

Now $z = g(x, y)$. This implies that

$$\frac{dz}{dt} = \frac{\partial g}{\partial x} \frac{dx}{dt} + \frac{\partial g}{\partial y} \frac{dy}{dt}.$$

Hence

$$\begin{aligned} LHS &= \int_a^b \left[(P + Rg_x) \frac{dx}{dt} + (Q + Rg_y) \frac{dy}{dt} \right] dt \\ &= \int_C (P + Rg_x) dx + (Q + Rg_y) dy \end{aligned}$$

By Green's theorem, the above integral becomes

$$= \int_D \int \left[\frac{\partial}{\partial x} (Q + Rg_y) - \frac{\partial}{\partial y} (P + Rg_x) \right] dA.$$

Let us concentrate on $\frac{\partial}{\partial x} (Q + Rg_y)$ first. Both Q and R are functions of x, y, z and z is again a function of x and y . So by Chain rule, we have

$$\frac{\partial}{\partial x} (Q + Rg_y) = (Q_x + Q_z z_x) + (R_x + R_z z_x) g_y + R \frac{\partial^2 z}{\partial x \partial y}.$$

Since $z = g(x, y)$, we get

$$\frac{\partial}{\partial x} (Q + Rg_y) = (Q_x + Q_z g_x) + (R_x + R_z g_x) g_y + R \frac{\partial^2 z}{\partial x \partial y}.$$

Similarly, we can get an expression for $\frac{\partial}{\partial y} (P + Rg_x)$. Once we subtract and simplify these expressions, we will get the RHS (*). $\square$

Remark 3.0.3. If we consider a globe, then we can think of it to be bounded by the equator circle as well as some other circle towards the top (nearer to the north pole). As we go up on the globe (towards the north pole) reducing the size of the circle, we will eventually get a point. These kind of concepts are going to be very very useful in physics, and important in working out some of the examples.

—————END—————